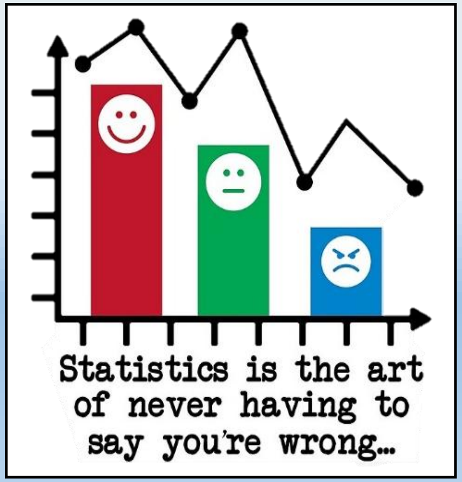
Mathematics II (part 1)

STATISTICS AND PROBABILITY



SINGAPORE INSTITUTE OF TECHNOLOGY

Ryan Fraser Kirwan

Mathematics II (part 1)

Module schedule Trimester 2

Wk	Topic
1	Descriptive Statistics
2	Probability Theory
3	Discrete Probability Distributions
4	Continuous Probability Distributions
5	Continuous and Sampling Distributions
6	Continuation and Combination of topics + Revision sessions
7	Recess Week
8	Linear Equations and Matrices
9	Vectors and Geometry
10	Linear Transformations/Regression
11	Linear Systems + Homogeneous Systems and Null Spaces
12	Matrix Decomposition + Eigenvalues
13	Singular Value Decomposition (SVD) + Application of SVD to Machine Learning
14	Exam Week
16-18	Trimester Break

Attention: These topics are tentative, so may be subject to changes. Also, the final weeks will be very busy, so please plan appropriately. Thank you

Mathematics II (part 1)

MODULE INTRODUCTION: Team Based Learning

Revision with possible consultation
10 mins

→

LMS iRAT with Respondus browser
20 mins

→

Team revision and discussion in groups
20 mins

→

LMS tRAT with Respondus (but done as individuals)
10 mins

→

Discussion of solutions in lecture format (clarifications)
40 mins

BREAK
~10 mins

16:00 -> Pre-quiz chat/Consultation

16:10-16:20 -> begin iRAT (10mins for enter/exiting quiz, 19+1min for quiz)

16:40-17:00 -> rejoin main Zoom, then go into break-out rooms (20mins)

17:00-17:10 -> begin tRAT (10mins for enter/exiting quiz, 9+1min for quiz)

17:20 -> presentation of Solutions Lecture (runs until end of session)

SINGAPORE INSTITUTE OF TECHNOLOGY

Mathematics II (part 1)

MODULE INTRODUCTION: RATs

Each Week:

- Allocate Lecture Content and Exercise questions to complete
  - Deal with new content in Tutorial Sessions (week prior to assessment)
- RAT questions to cover Lecture content, Tutorials and **Video Resources**.
- RATs to be ~10 questions of MCQ and short-answer (number input for calculations)
- iRATs and tRATs for Weeks 2-6 (Assessments 1-5)
  - 25% (5% total each: 3% iRAT, 2% tRAT)
- Exam 50% for Week 14 (of this, half will be on first half content i.e., worth 25%)
  - Ideally this will be a proctored exam (done on paper), but this is pending feasibility

Enhanced video content + additional Stats Resources on the LMS

SINGAPORE INSTITUTE OF TECHNOLOGY

Mathematics II (part 1)

(optional) Resources

- **OpenIntro Statistics 2nd Edition**
  by David Diez, Christopher Barr and Mine Cetinkaya-Rundel
  http://www.openintro.org/stat/textbook.php?stat_book=os

- Mind on Statistics 5th Ed. by J. M. Utts and R. F. Heckard Cengage Learning, 2015

- Introduction to Probability and Statistics 14th Ed. by W. Mendenhall, R. J. Beaver and B. M. Beaver Cengage Learning, 2013

STATISTICS

INTRODUCTION TO STATISTICS

What is Statistics?

- Our world is full of uncertainties and our life is filled with questions?

Should I pursue Computing degree programme?

Is using mobile phones harmful to health?

Is global warming propaganda?

- Statistics enables us to collect and discover patterns in data so as to predict and make wiser decisions for our questions.

Introduction to Statistics

Types of Statistics

Descriptive

- Collecting
- Organizing
- Summarizing
- Presenting data

Inferential

- Making inferences
- Hypothesis testing
- Making predictions
- Determining relationships

Types of Statistics: Example 1

- Of 280 randomly selected people in a town in Italy, 210 people had the last name Nicolussi.
- An example of descriptive statistics is the following statement:

"210 (75%) of these 280 people have the last name Nicolussi."

- An example of inferential statistics could be the following statement :

"75% of all people living in Italy have the last name Nicolussi."

Types of Statistics : Example 2

- On the last 3 Sundays, Henry sold 2, 1, and 0 new cars respectively.
- An example of descriptive statistics is the following statement :

  "Henry averaged 1 new car sold, per Sunday, for the last 3 Sundays."

- An example of inferential statistics is the following statement :

  "Henry never sells more than 2 cars on a Sunday."

Types of Statistics

Do you think the following are Descriptive or Inferential?

[ZOOM POLL]

Descriptive or Inferential ?

1. Eighty-two percent of the 100 employees from a small local company attended the annual company picnic. (Descriptive)

2. The average age of freshman at the University of Georgia is currently 20.8 years old, based on the information from the registrar's office. (Descriptive)

3. The average number of hours vacationers spend in American national parks during the summer months is 4.5 hours. (Inferential)

Types of Statistics

Descriptive

- Collecting
- Organizing
- Summarizing
- Presenting Data

Probability Theory

"Probability Theory is the basis to link Descriptive Statistics with Inferential Statistics."

Inferential

- Making Inference
- Determining relationships
- Making predictions
- Hypothesis Testing

Introduction to Statistics

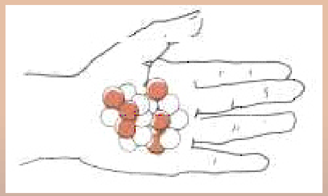
Types of Statistics

Descriptive	Probability Theory	Inferential
• Collecting • Organizing • Summarizing • Presenting Data	"Probability Theory is the basis to link Descriptive Statistics with Inferential Statistics."	• Making Inference • Determining relationships • Making predictions • Hypothesis Testing

Descriptive

Turning data into information

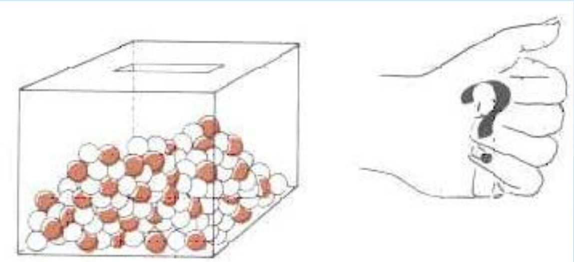
- Organize, summarize, and describe the data (outcomes from sample data).



Probability Theory

Understanding the uncertainty in data

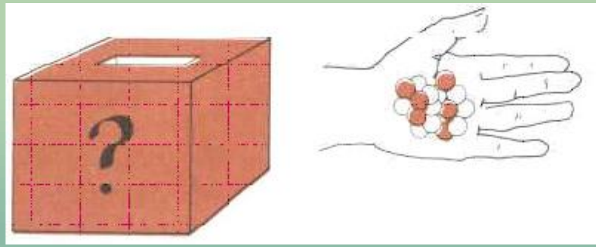
- Given the population, and sample size then predict applicability of the outcomes.



Inferential

Making decision from data

- Given the outcomes, we can infer for the population in order to make decisions.



Types of Statistics

The learning outcomes for Statistics are…

✓Turning Data into Information in life

- Use descriptive statistical techniques to organize, summarize and present data with graphical and numerical measures.

✓Understanding Uncertainty in life

- Apply probability theory and distribution models to calculate probabilities of real-world events.

✓Making Decisions from Data in life

- Conduct inferential statistical analysis and testing to draw conclusions from data.

Ambiguous or misleading comparisons

- Bank A: "Our interest rates are 1% lower"
  - *Ambiguity 1: Lower than which Bank?*
  - *Ambiguity 2: If the Bank B offers 2% interest rates, does Bank A offer 1% or 1.98% ?*

- Misleading Tweets (2019)
  - *" President Trump's approval rating among Latinos going (up) to 50%," Trump tweeted.*
  - *National polling results for only 153 Latino voters (margin of error is ±9.9%):*

"I can prove it or disprove it! What do you want me to do?"

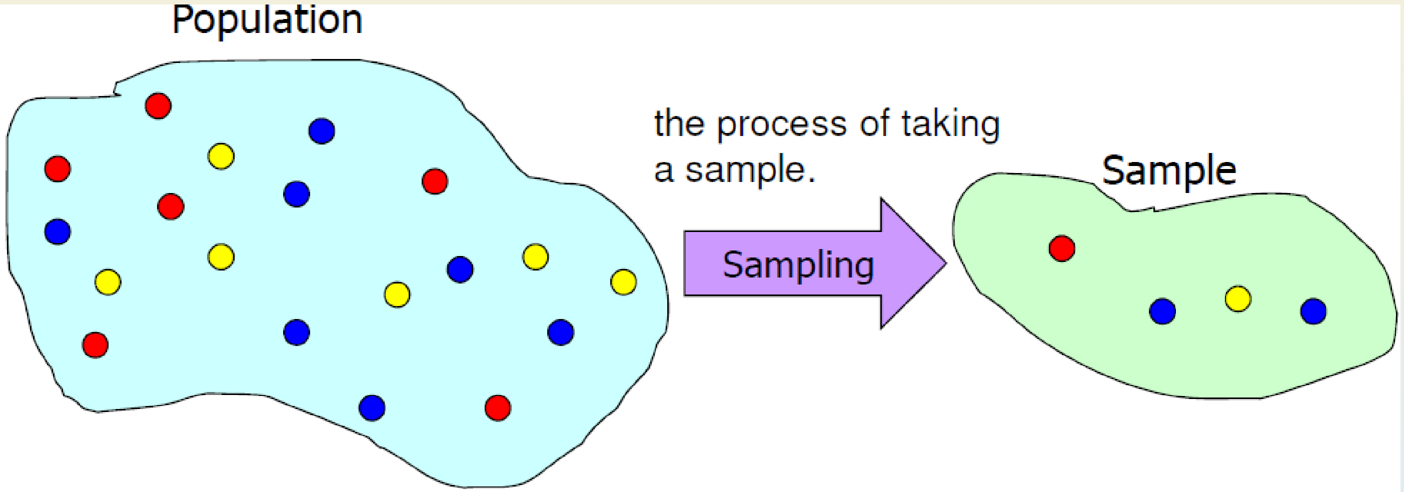
DESCRIPTIVE STATISTICS

DATA AND SAMPLES

Data and Samples

Population versus Sample

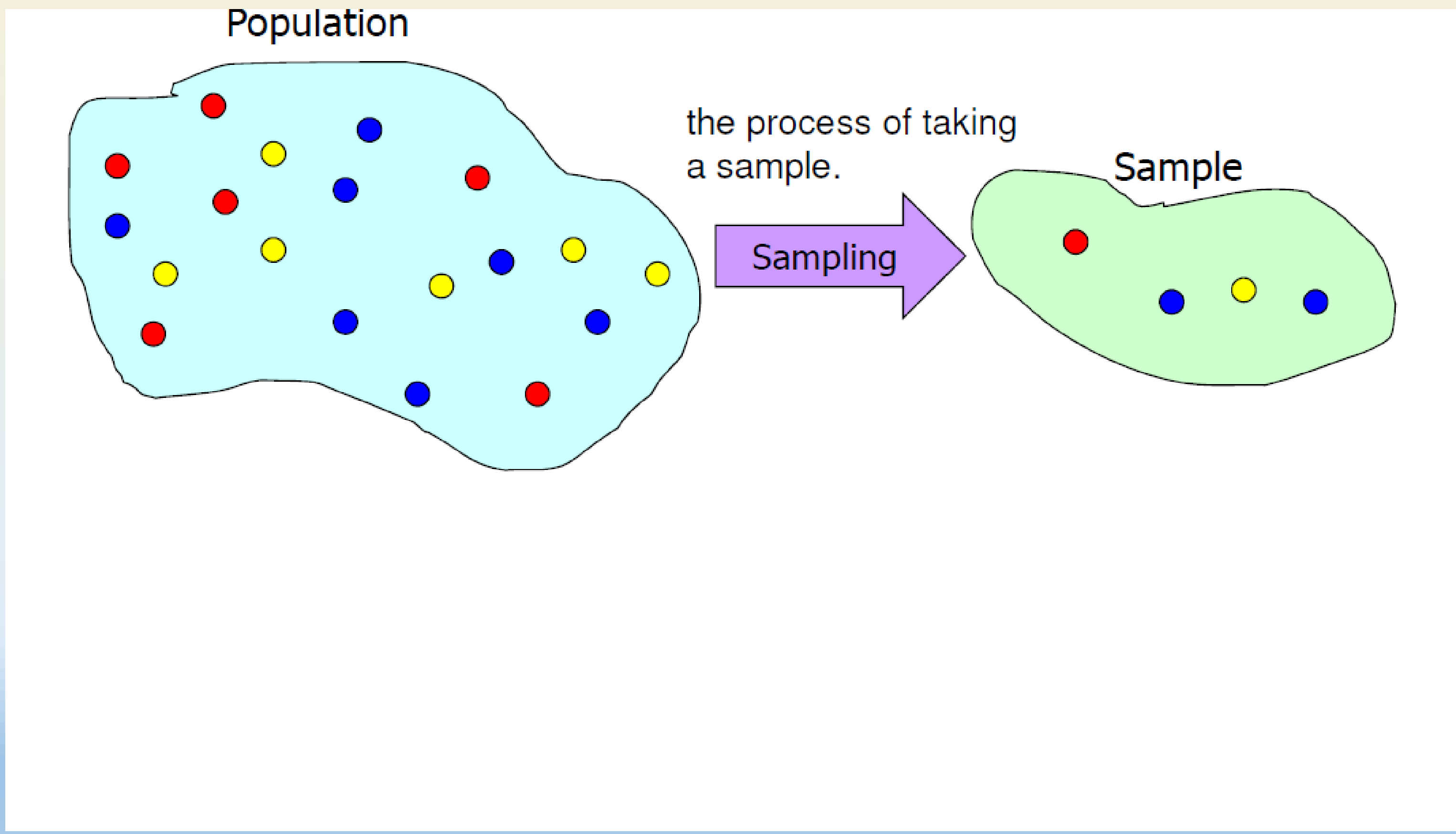
To be accurate, we need to conduct census to collect all the data. But to be practical, sampling is commonly done instead.



- Population
  - Census – whole population which is usually large, is being measured, which can be costly and may be impractical/impossible.
- Sample
  - Sampling – a sample, which is a small subset of the population is measured (which is then used to **infer** the population).

Population versus Sample

The data collected from the population or sample are then used to compute the population parameter or sample statistic respectively.



Important to know what the symbols represent

DESCRIPTIVE STATISTICS

TYPES OF DATA AND DATA PRESENTATION

Types of Data

- In Statistics, it is important to differentiate data/variables as qualitative (categorical or non-numeric) or quantitative (numeric) because they are treated differently.

Expressed numerically. Can be used in calculations

Data

Describe a characteristic or quality of variable

Quantitative (numeric)

scale for measurement has numerical values that represent different magnitudes of the variable

Qualitative (Categorical)

scale for measurement is a set of categories

Quantitative/Numerical variable

(scale for measurement is numerical values that differ in magnitude)

Examples:

- Age, height, weight, BMI = weight(kg)/[height(m)]^2
- GPA
- Time spent on Internet yesterday
- Reaction time to a stimulus

Qualitative/Categorical variable

(scale for measurement is a set of categories)

Examples:

- Racial-ethnic group (Asian, White, Black, Latino)
- Political party identification (Labour, Conservative, Liberal-democrats)
- Vegetarian? (yes, no)
- Mental health evaluation (well, mild symptom formation, moderate symptom formation, impaired)
- Happiness (very happy, pretty happy, not too happy)

Types of Data and Data Presentation

Types of Data

- In Statistics, it is important to differentiate data/variables as qualitative (categorical or non-numeric) or quantitative (numeric) because they are treated differently.

Data

Quantitative (numeric)

Qualitative (Categorical)

Discrete (countable)

- Possible values form a set of separate numbers

Continuous (Non-countable)

- Possible values can take an infinite continuum of possible real number values

[ZOOM POLL]

Types of Data

Which of the following is an example of a Qualitative or Categorical variable (data)?

[ZOOM POLL]

- Height in inches
- Zip code
- Years of schooling completed
- Annual income in dollars

Which of the following is an example of a Quantitative or Numerical variable (data)?

- Religious affiliation
- University major (degree)
- Number of life-events in the past year
- High school graduate or not

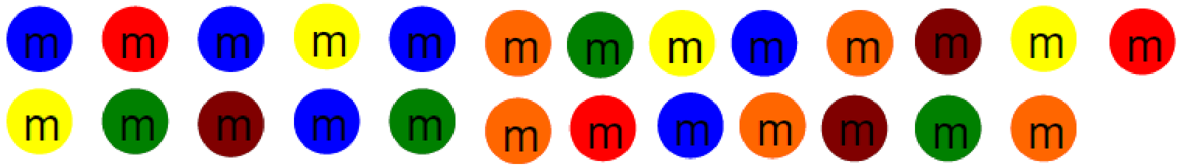
Which of the following is NOT an example of a Continuous variable (data)?

- Annual income
- Number of family members
- Finish time for a marathon
- Distance travelled to work

Representing Categorical Data

- How would you represent these data?
  (a set of colored M&Ms)
- To obtain information from data, qualitative (categorical) data are often summarized in a table using frequency or relative frequency.

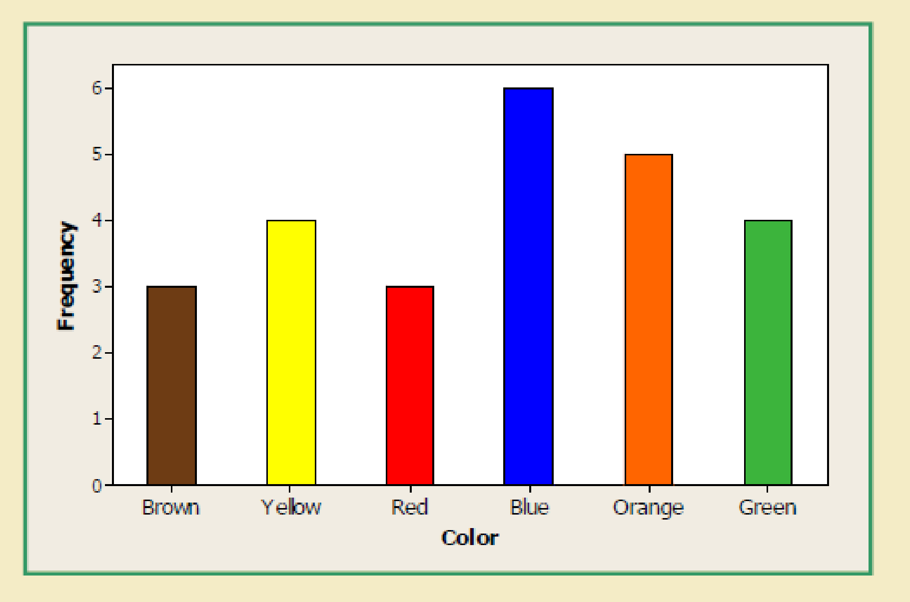
Data:



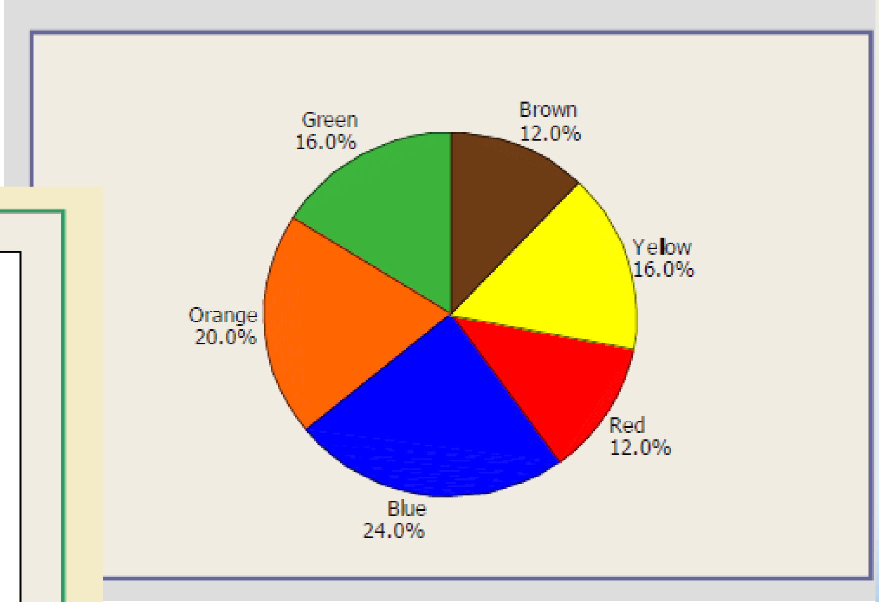
Representing Categorical Data

- Graphically, qualitative (categorical) data are commonly presented using bar chart or pie chart.

Bar chart:



Pie chart:



Representing Categorical Data

- The collection of all categories (classes) and the corresponding relative frequencies is called the relative frequency distribution of the variable.

Definition: Population proportion or relative frequency is defined as:

$$\boxed{p_i = \frac{f_i}{N}}$$

Similarly, sample proportion:

$$\boxed{\hat{p}_i = \frac{f_i}{n}}$$

where $f_i =$ frequency

$N = \sum f_i$ (all data)

$n =$ a subset of N

8

Color	Relative Frequency
Red	0.12
Blue	0.24
Green	0.16
Orange	0.20
Brown	0.12
Yellow	0.16

Representing Quantitative Data

- Numeric data can be identified as frequency distribution as well.

Data:

Daily Number of Internet System Crashes

1	3	1	1	0	1	0	1	1	0
2	2	0	0	0	1	2	1	2	0
0	1	6	4	3	3	1	2	4	0

Frequency distribution:

Value x	Frequency	Relative Frequency
0	9	.300
1	10	.333
2	5	.167
3	3	.100
4	2	.067
5	0	.000
6	1	.033
Total	30	1.000

Data

Quantitative (numeric)

Qualitative (Categorical)

Discrete (countable)

- Possible values form a set of separate numbers

Representing Quantitative Data

- Numeric data can be identified as frequency distribution as well.

Data:

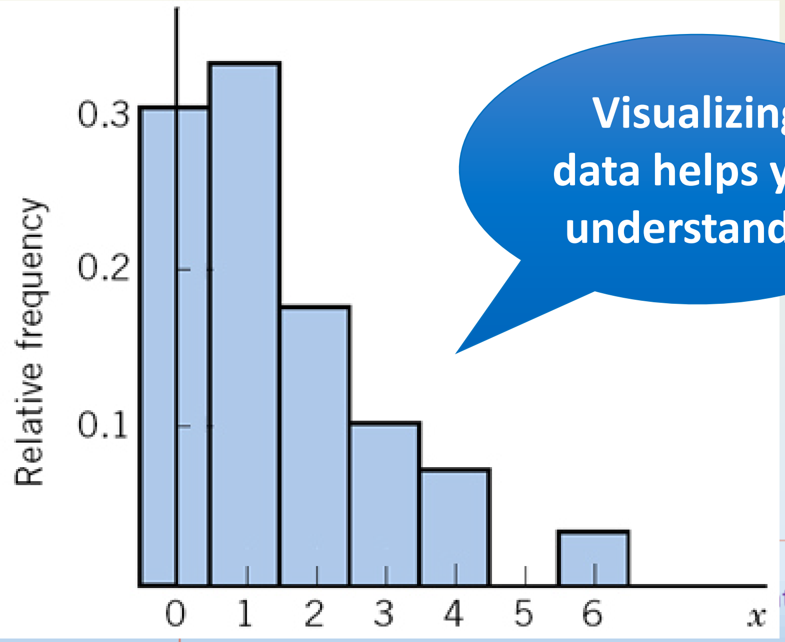
Daily Number of Internet System Crashes

1	3	1	1	0	1	0	1	1	0
2	2	0	0	0	1	2	1	2	0
0	1	6	4	3	3	1	2	4	0

Frequency distribution:

Value x	Frequency	Relative Frequency
0	9	.300
1	10	.333
2	5	.167
3	3	.100
4	2	.067
5	0	.000
6	1	.033
Total	30	1.000

However, quantitative data are commonly presented graphically using histograms instead.



Visualizing data helps you understand it

tegorical)

Discrete (countable)

- Possible values form a set of separate numbers

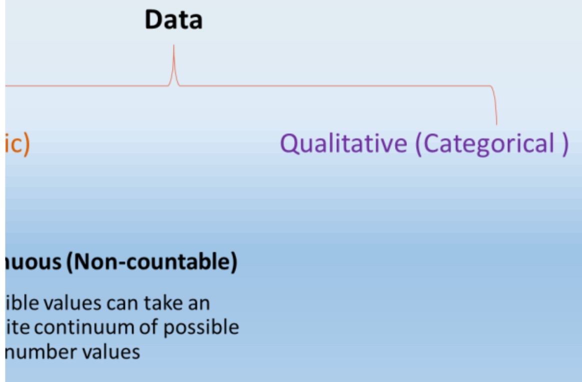
Types of Data and Data Presentation

Representing Large or Continuous Quantitative Data

- For large dataset and/or quantitative continuous data, **classes (covering continuous non-overlapping range of values)** are commonly used to group data together.

Data: A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	



Types of Data and Data Presentation

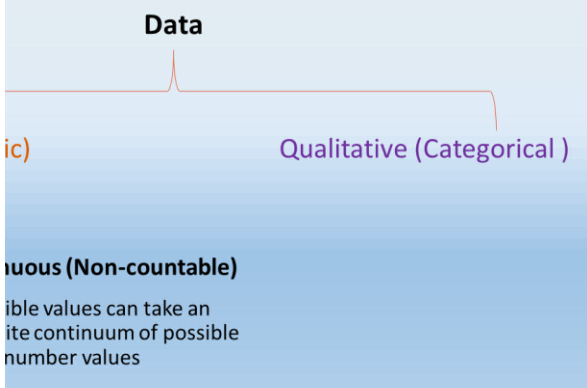
Representing Large or Continuous Quantitative Data

- For large dataset and/or quantitative continuous data, classes (covering continuous non-overlapping range of values) are commonly used to group data together.

Data: A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	

10 to < 13 (interval includes 10 but not 13)



Types of Data and Data Presentation

Representing Large or Continuous Quantitative Data

- For large dataset and/or quantitative continuous data, classes (covering continuous non-overlapping range of values) are commonly used to group data together.

Data:

A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	

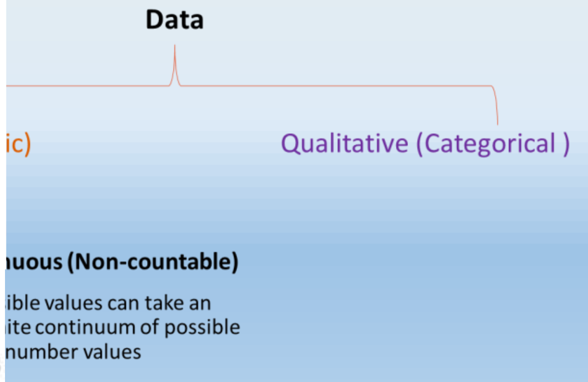
Frequency table:

Class
10 to < 13
13 to < 16
16 to < 19
19 to < 22
22 to < 25
25 to < 28
28 to < 31

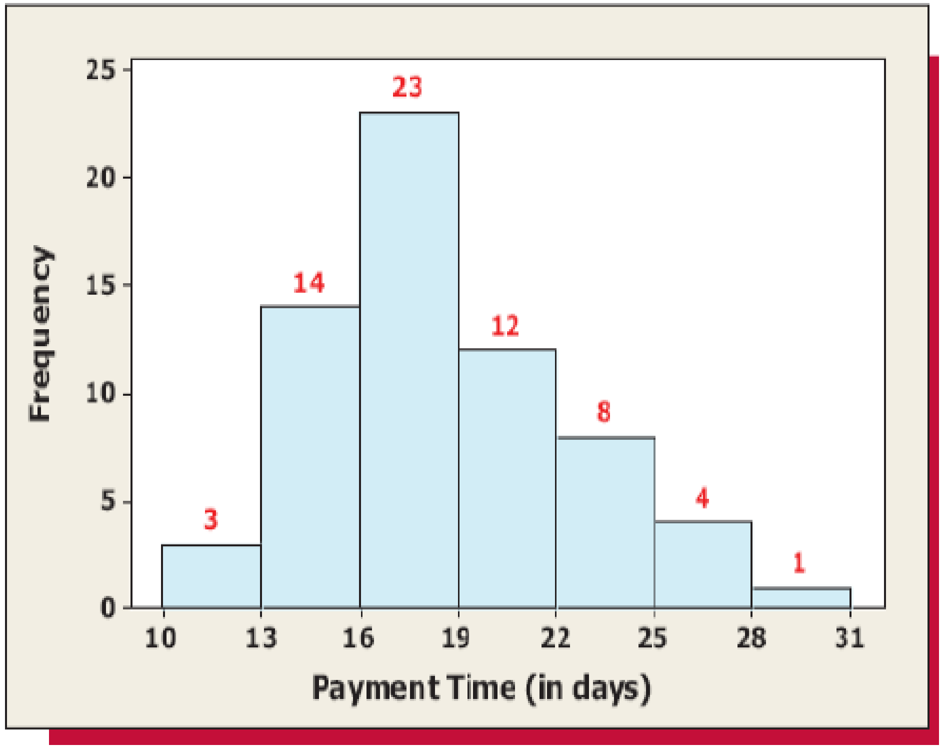
10 to < 13 (interval includes 10 but not 13)

Note: Use enough classes (usually 5-14) to show variation in data, but not too many classes that many contain only a few data items.

13



Data Presentation: Histograms

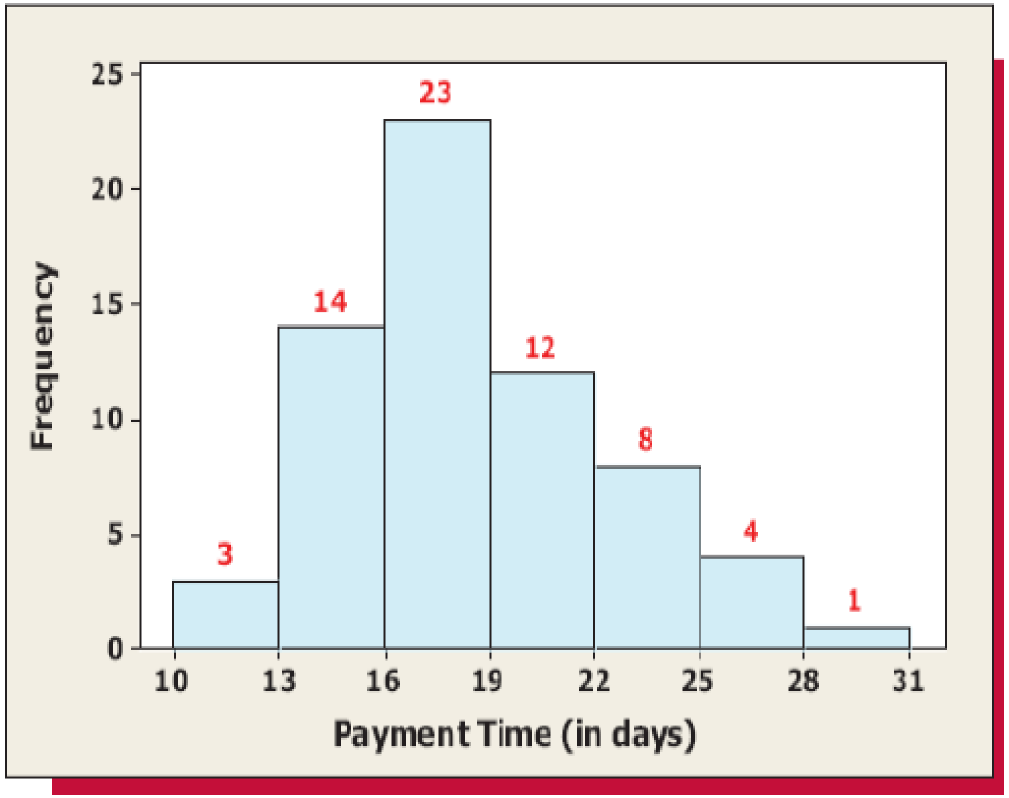


Frequency table:

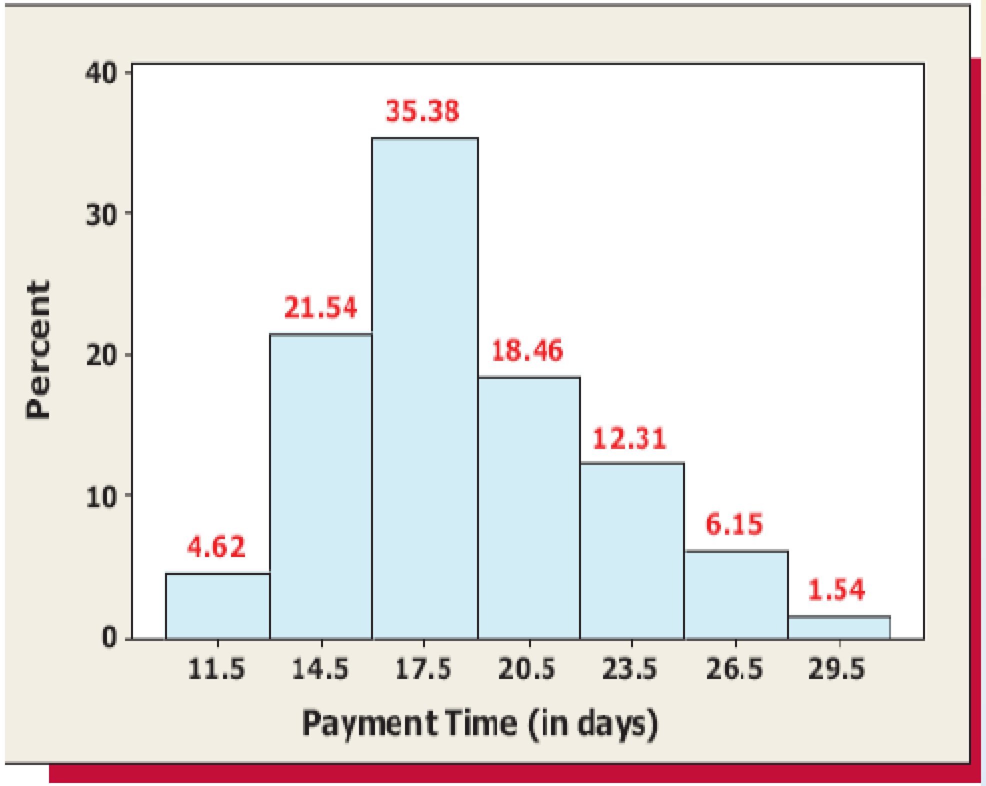
Class	All 65 Tally Marks	Frequency
10 **to** < 13	III	3
13 **to** < 16	~~IIII~~ ~~IIII~~ IIII	14
16 **to** < 19	~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ III	23
19 **to** < 22	~~IIII~~ ~~IIII~~ II	12
22 **to** < 25	~~IIII~~ III	8
25 **to** < 28	IIII	4
28 **to** < 31	I	1

Types of Data and Data Presentation

Data Presentation: Histograms



(a)



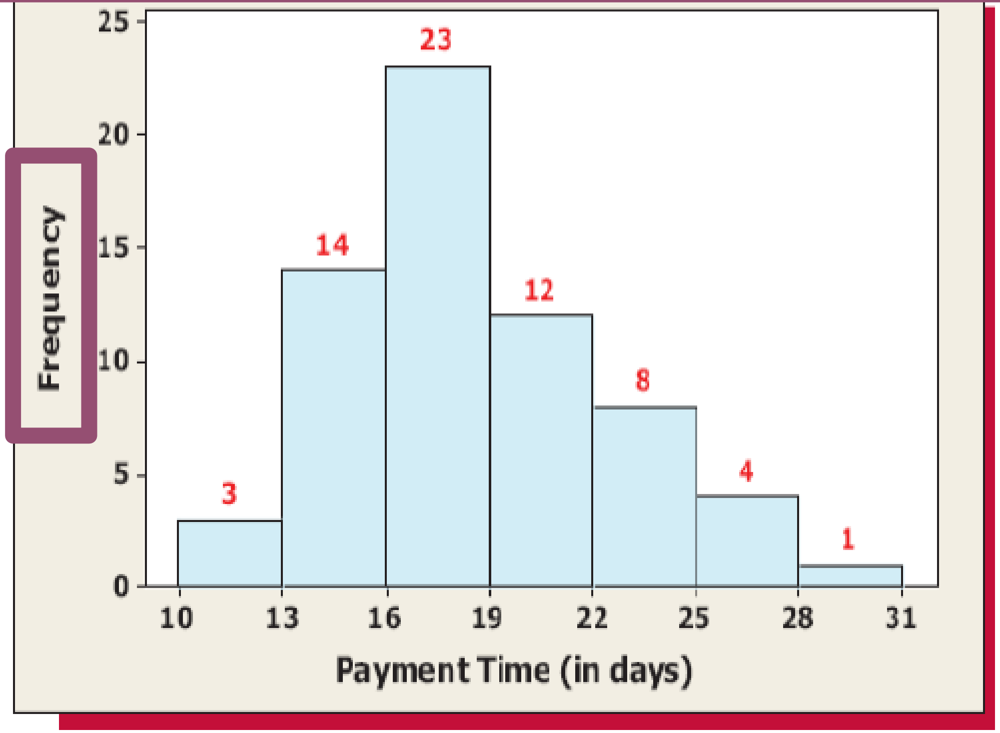
(b)

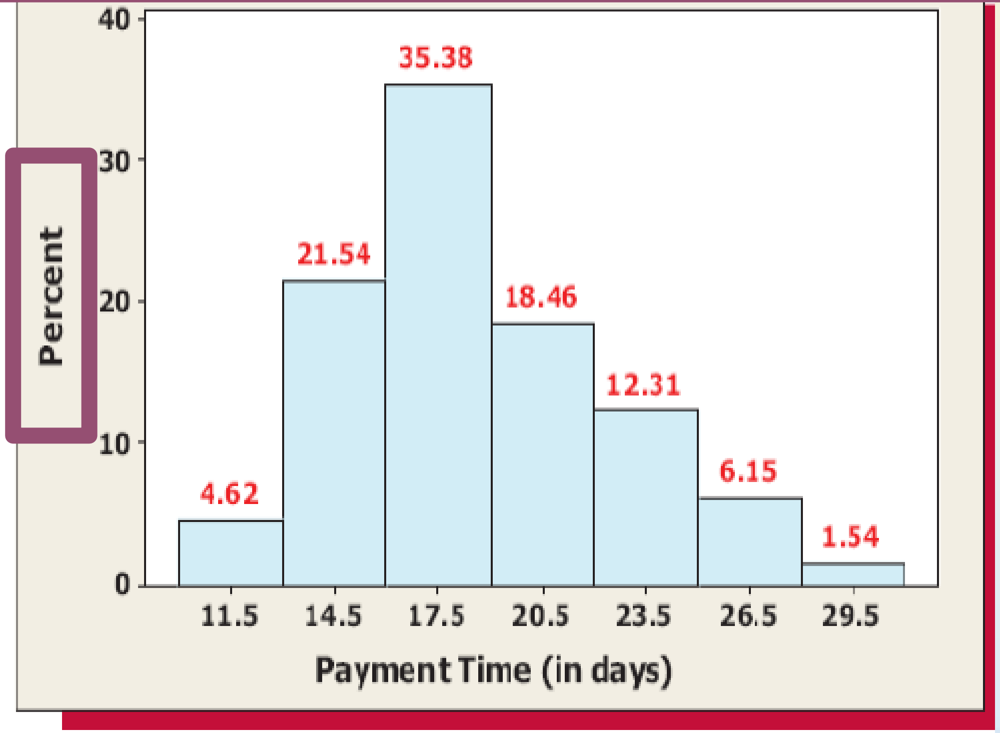
These two diagrams represent same data, spot the differences

Spot the Difference

Data Presentation: Histograms

Graphically, quantitative continuous data or grouped data are also commonly presented using histograms

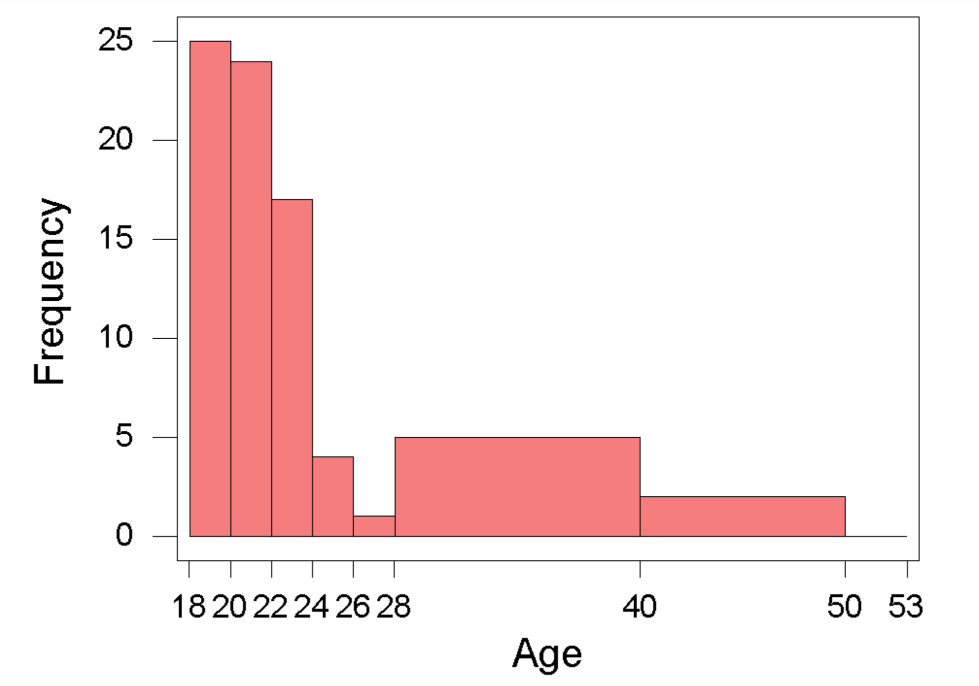




Note that there is no gap between the rectangles in the histogram because the classes cover continuous non-overlapping range.

Data Presentation: Histograms with Unequal Classes

- Sometimes, it may be desirable to group data into unequal class width.
- However, for unequal class width, the corresponding histogram can be misleading if it is simply drawn using frequency or relative frequency.
- E.g. the ages of 79 undergraduate students in a university



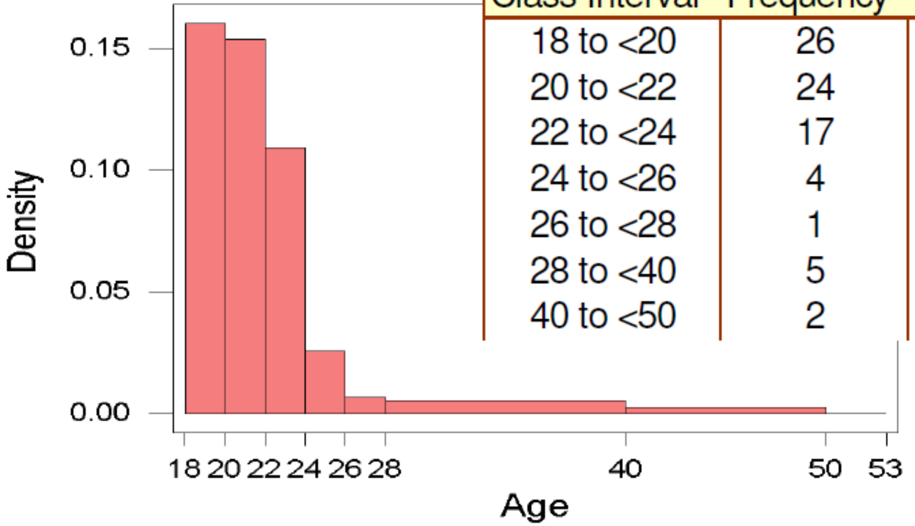
Class Interval	Frequency	Relative Frequency
18 to <20	26	0.329
20 to <22	24	0.304
22 to <24	17	0.215
24 to <26	4	0.051
26 to <28	1	0.013
28 to <40	5	0.063
40 to <50	2	0.025

Data Presentation: Histograms with Unequal Classes

- E.g. the ages of 79 undergraduate students in a university
- To avoid misinterpretation, histogram needs to be drawn using density and not frequency, when presenting continuous data or grouped data with unequal class width.

$$\text{density} = \frac{\text{relative frequency}}{\text{class width}}$$

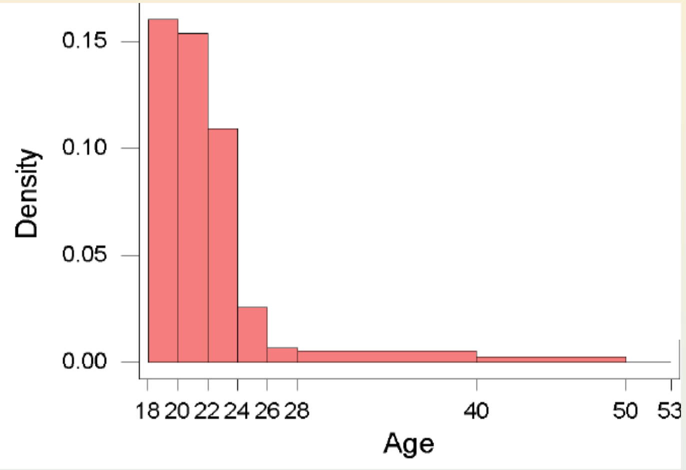
Class Interval	Frequency	Relative Frequency	Density
18 to <20	26	0.329	0.165
20 to <22	24	0.304	0.152
22 to <24	17	0.215	0.108
24 to <26	4	0.051	0.026
26 to <28	1	0.013	0.007
28 to <40	5	0.063	0.005
40 to <50	2	0.025	0.003



Types of Data and Data Presentation

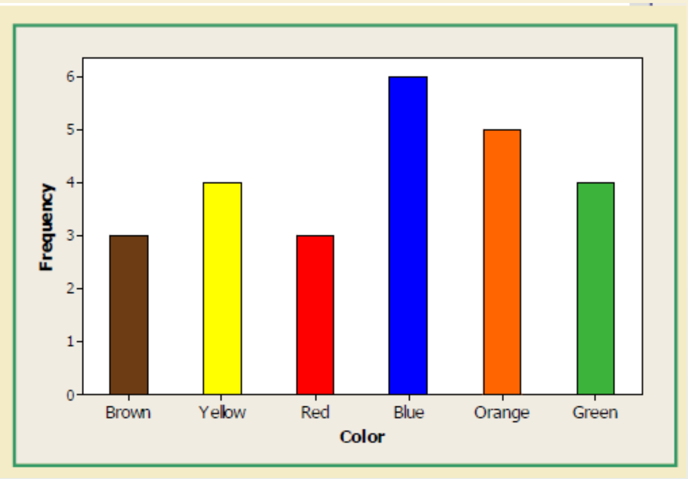
Data Presentation: Histograms with Unequal Classes

(density) Histogram



- Quantitative
- Bars can not be reordered
- Bar width represents bins, classes or intervals of values – can be different in length
- No gaps (if there are gaps, they are value zero)
- Area of the bar denotes the frequency, not height
- Histogram show distributions

Bar chart



- Categorical
- Bars can be reordered
- Bar width is the same, the width has no meaning
- There are gaps (no value to gaps, just space)
- The height of the bar is the frequency, the width has no meaning
- Bar chart compare variables

Bivariate data

- What if we wish to investigate the relationship between two variables and summarize data?
  - e.g. quality rating and meal price for 300 restaurants:
    - What can we make of such data?
    - Is an expensive meal always excellent?
    - How much is an expensive meal?
    - Are there any excellent cheap meals?

	variable 1	variable 2
Restaurant	**Quality Rating**	**Meal Price ($)**
1	Good	18
2	Very Good	22
3	Good	28
4	Excellent	38
5	Very Good	33
6	Good	28
7	Very Good	19
8	Very Good	11
9	Very Good	23
10	Good	13
.	.	.
.	.	.
.	.	.

Types of Data and Data Presentation

Bivariate data

Crosstabulation (or crosstab) is a method for summarizing data for two variables in a table called a ***contingency table.***

variable 1 → Quality Rating; variable 2 → Meal Price ($)

Restaurant	Quality Rating	Meal Price ($)
1	Good	18
2	Very Good	22
3	Good	28
4	Excellent	38
5	Very Good	33
6	Good	28
7	Very Good	19
8	Very Good	11
9	Very Good	23
		13
.	.	.
.	.	.
.	.	.

	Meal Price				
Quality Rating	$10–19	$20–29	$30–39	$40–49	Total

Types of Data and Data Presentation

Crosstabulation using Contingency Tables

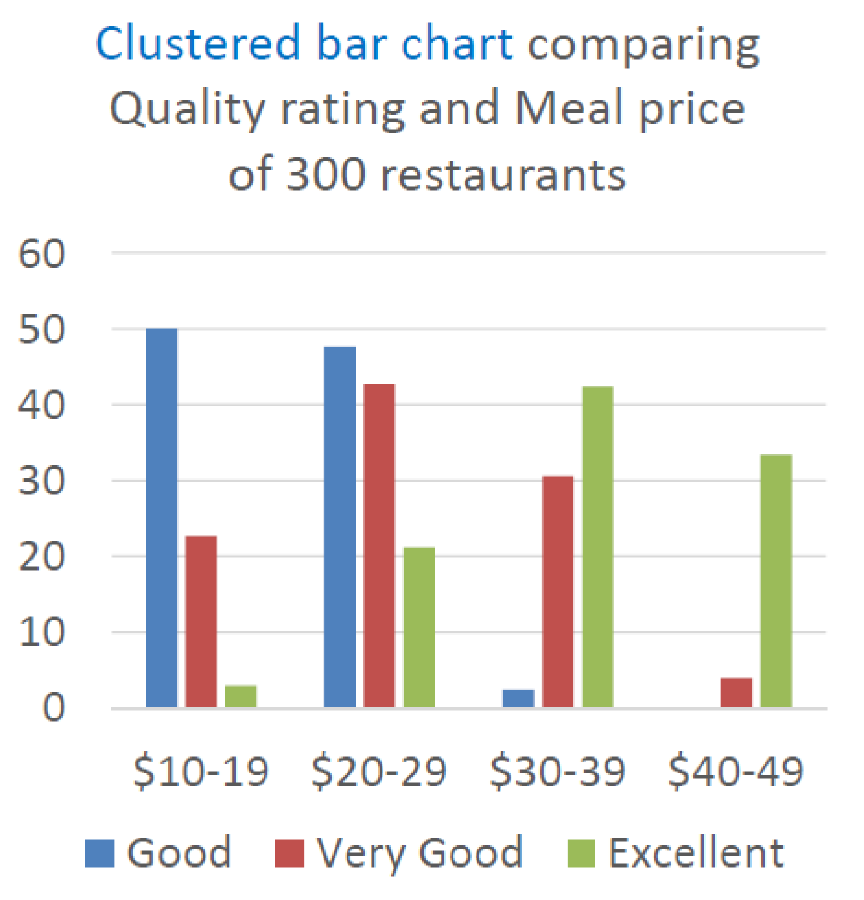
Crosstab can be used for both qualitative data, both quantitative data, or one qualitative vs one quantitative data.

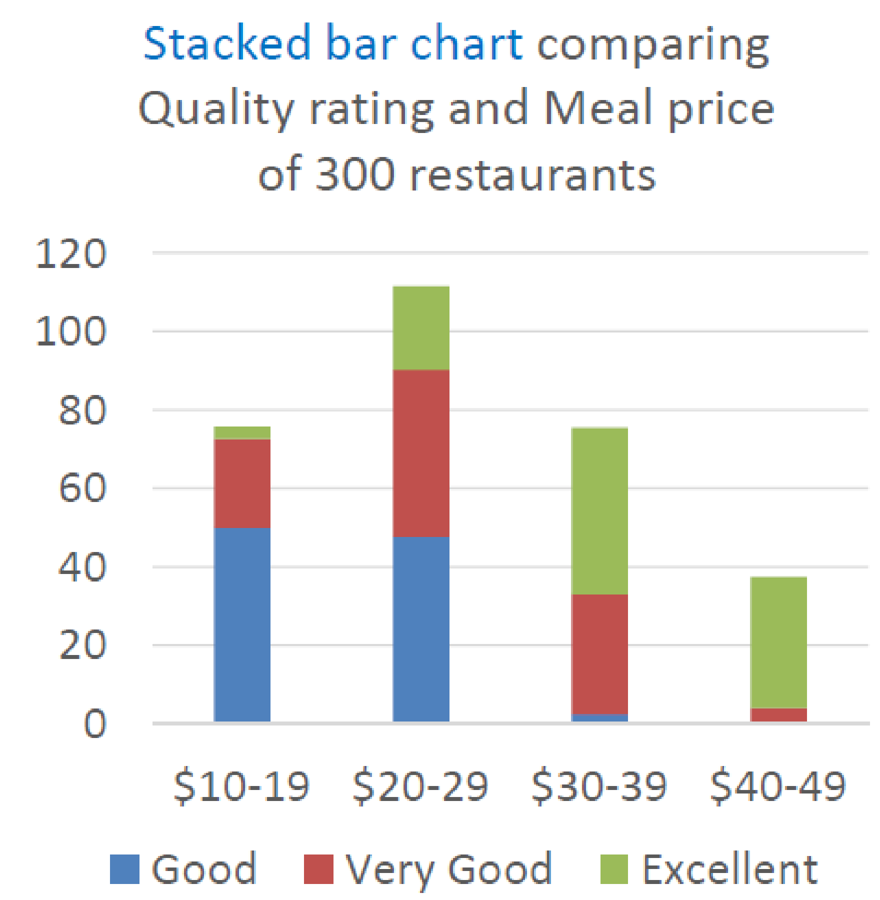
	Meal Price				
Quality Rating	**$10–19**	**$20–29**	**$30–39**	**$40–49**	**Total**
Good	42	40	2	0	84
Very Good	34	64	46	6	150
Excellent	2	14	28	22	66
Total	78	118	76	28	300

	Meal Price				
Quality Rating	**$10–19**	**$20–29**	**$30–39**	**$40–49**	**Total**

Crosstabulation using Clustered/Stacked bar chart

- For easier comparison, cross-tabulation may be presented graphically using clustered bar chart or stacked bar chart.





Summarizing Quantitative Data

- *Imagine you were living in the olden days when Statistics was being developed and you were asked to propose a* ***SINGLE number*** *to represent a data set, what would you propose?*

- *…enter Measure of Location!*

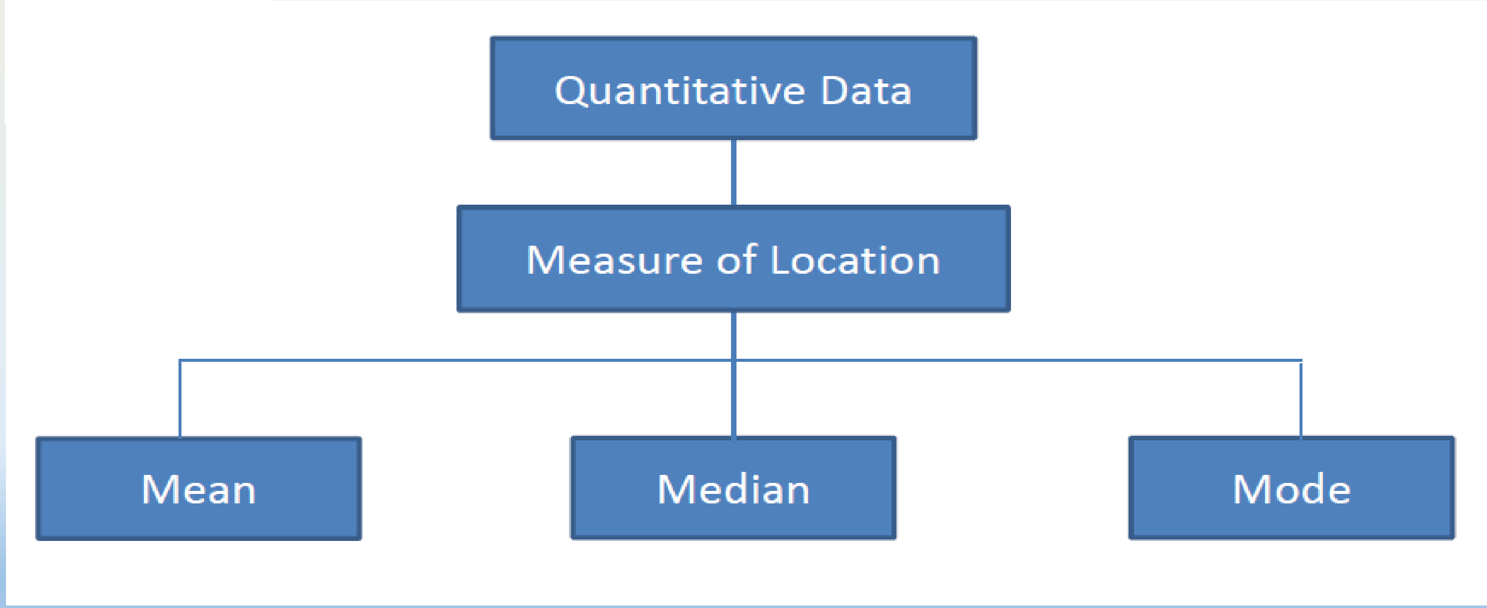
DESCRIPTIVE STATISTICS

MEASURE OF LOCATION

Measure of Location

Summarizing Quantitative Data

- Numerical measures can also be used to summarize data. An example is the *measure of location*.
- Mathematicians have defined 3 measures of location that are widely used:



Mean

- Mean is defined as the numerical average, calculated as the sum of the data values divided by the number of values.

Definition: Population mean, denoted as μ , is defined as:

$$\boxed{\mu = \frac{x_1 + x_2 + \ldots + x_N}{N} = \frac{\sum x_i}{N}}$$

where $x_1, x_2, \ldots, x_N$ are ALL the N data in the population.

Similarly, sample mean, denoted as $\overline{x}$, is defined as:

$$\boxed{\overline{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{\sum x_i}{n}}$$

where $x_1, x_2, \ldots, x_n$ is a subset of $n < N$ data in the population.

22

Mean: Example

Example: The numbers of stray dogs captured and turned in to a town animal shelter on the last 20 days were:

4	6	8	4	2	6	4	3	4	9
5	8	5	3	5	7	6	3	8	6

Find the mean.

(Mean for a day that is.)

Mean: Example

Example: The numbers of stray dogs captured and turned in to a town animal shelter on the last 20 days were:

4	6	8	4	2	6	4	3	4	9
5	8	5	3	5	7	6	3	8	6

Find the mean.

Solution:

Data x_i	Frequency f_i
2	1
3	3
4	4
5	3
6	4
7	1
8	3
9	1

$$= \frac{2\times1+3\times3+4\times4+\ldots8\times3+9\times1}{20}$$

$$= \frac{106}{20} = 5.3$$

$$\bar{x} = \frac{\sum x_i f_i}{n}$$

Mean: Grouped Data Example

How would you approach calculating the mean for grouped data?

Age	Frequency f_i
25 to < 33	5
33 to < 41	14
41 to < 49	13
49 to < 57	9
57 to < 65	7
65 to < 73	2

Measure of Location

Mean: Grouped Data Example

Age	Class Mark x_i	Frequency f_i
25 to < 33	(25 + 33)/2 = 29	5
33 to < 41	37	14
41 to < 49	45	13
49 to < 57	53	9
57 to < 65	61	7
65 to < 73	69	2

$$\bar{x} = \frac{\sum x_i f_i}{n}$$

$$= \frac{29\times5+37\times14+\ldots+69\times2}{50} = 45.8$$

(Get the average of each Age category.)

Measure of Location

Median

- Median is defined as the middle data value when the dataset has been ordered from the smallest to largest.

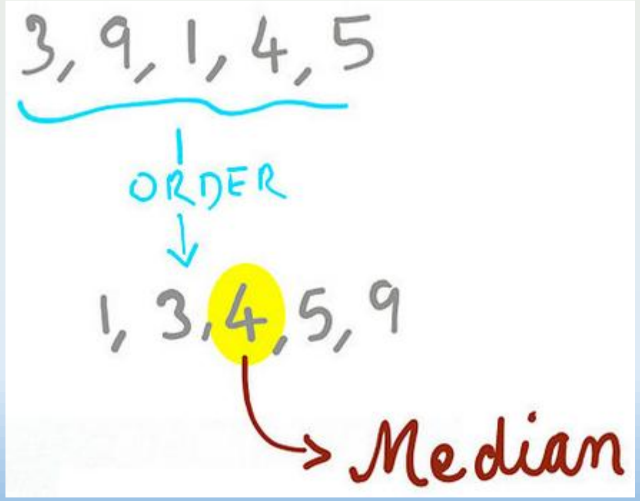
- What is the median of this data? 3, 9, 1, 4, 5

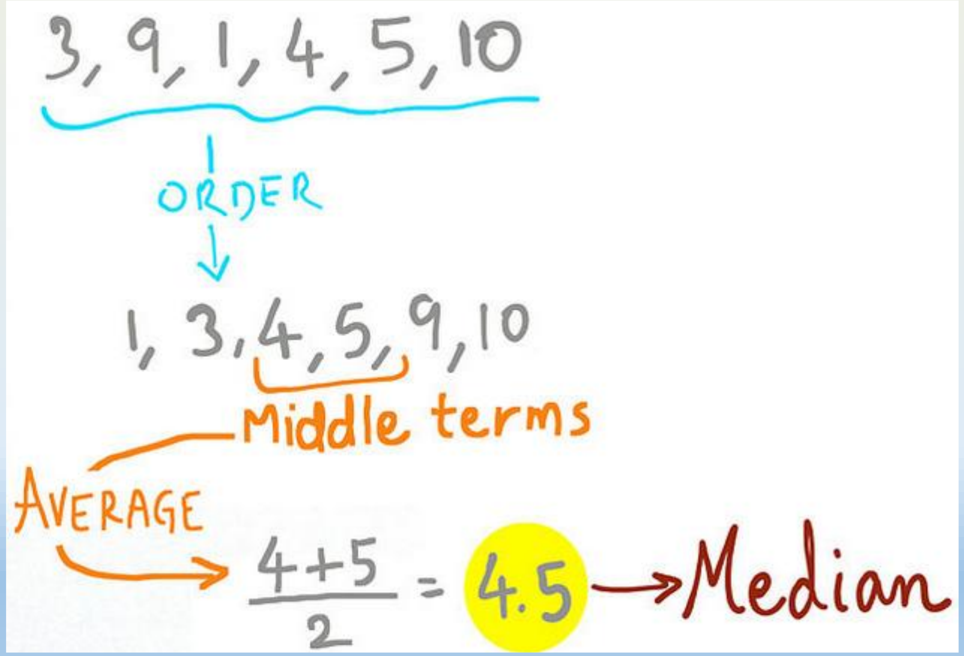
Measure of Location

Median: Example

- Median is defined as the middle data value when the dataset has been ordered from the smallest to largest.

- What is the median of this data?





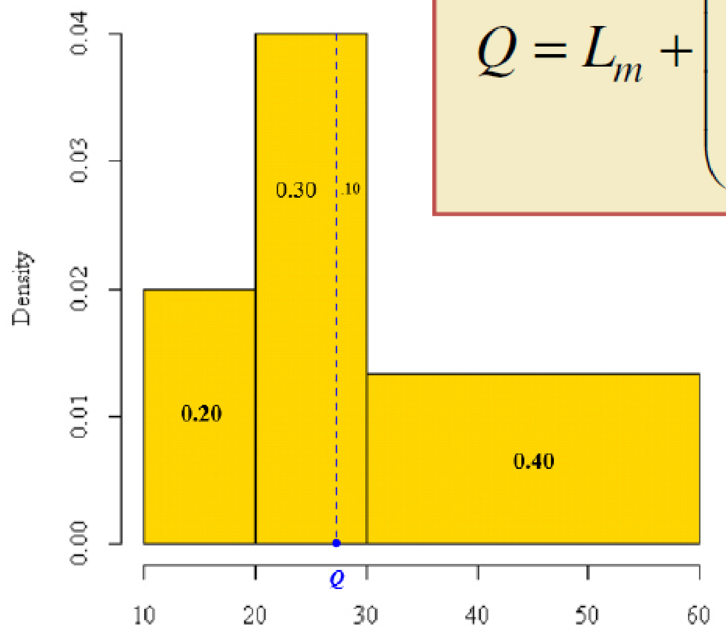
Median: Grouped Data Example

- For grouped data, median Q is an interpolated value within the class containing the median:

Class Interval	Frequency f_i	Relative Frequency $\frac{f_i}{n}$	Density (Rel Freq ÷ Class Width)
[10, 20)	**4**	0.20	0.02
[20, 30)	**8**	0.40	0.04
[30, 60)	**8**	0.40	0.013

Median: Grouped Data Example

- Median Q is an interpolated value within the class containing the median computed as follows:



$$Q = L_m + \left(\frac{\frac{n}{2} - cf_{m-1}}{f_m} \right) w_m = 20 + \left(\frac{\frac{20}{2} - 4}{8} \right)(10)$$

$$= 27.5$$

n : number of data

L_m : lower bound of class containing median

f_m : frequency of class containing median

w_m : width of class containing median

cf_{m-1} : cumulative frequency before the class containing median

27

Class Interval	Frequency f_i	Relative Frequency $\frac{f_i}{n}$	Density (Rel Freq ÷ Class Width)
[10, 20)	**4**	0.20	0.02
[20, 30)	**8**	0.40	0.04
[30, 60)	**8**	0.40	0.013

Measure of Location

Mode

- Mode is defined as the data value that has the highest frequency.
- What is the mode of the following data?

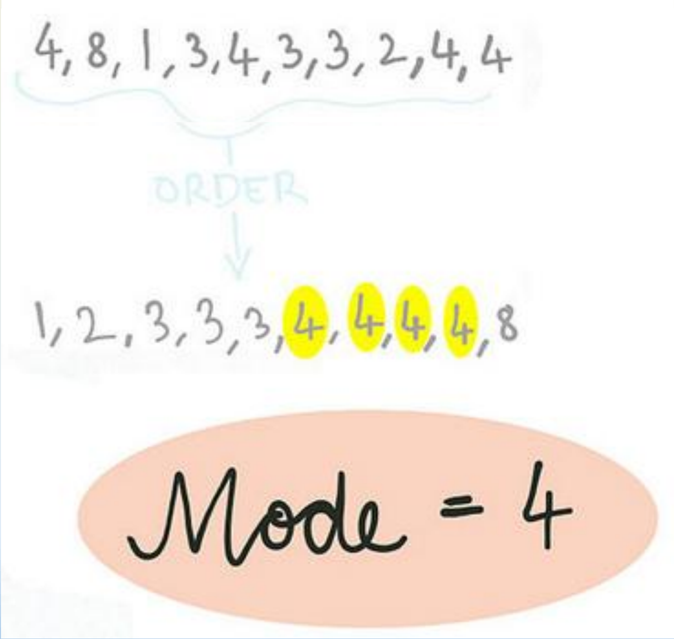
4, 8, 1, 3, 4, 3, 3, 2, 4, 4

Measure of Location

Mode: Example

- Mode is defined as the data value that has the highest frequency.

- For individual data



For grouped data:

Number of Cappuccinos	Freq
0 - 3	2
4 - 7	3
8 – 11	8
12 - 15	3
16 - 19	2

The class which has the highest frequency is called the modal class; i.e. 8 – 11

[ZOOM POLL]

Note: Mode is the only measure of location that conceptually can also be applied to qualitative data.

Mode: Example

What is the Mean of the following numbers: 3, 4, 8, 5, 2, 9, 11?

[ZOOM POLL]

- 4
- 5
- 6
- 7

What is the Median of the following numbers: 5, 9, 8, 10, 13, 4, 6?

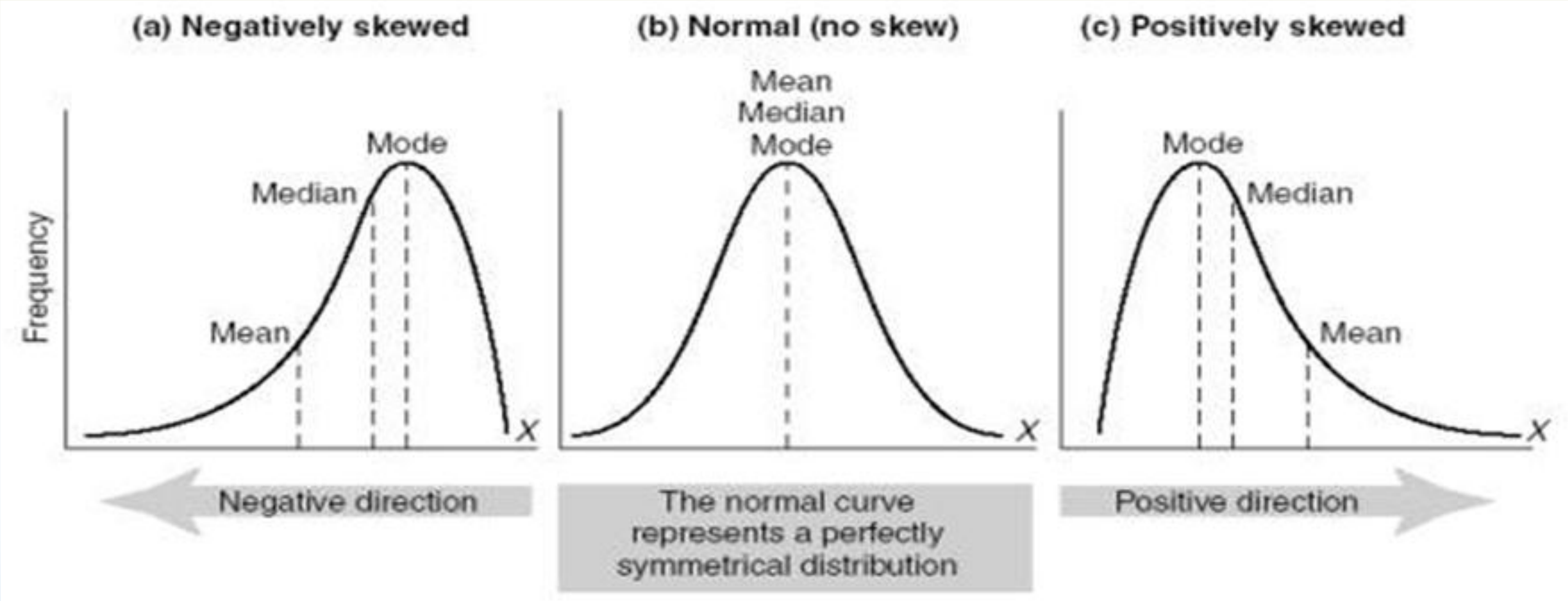
- 6
- 7
- 8
- 9

What is the Mode of the following numbers: 2, 3, 5, 3, 6, 7, 3, 2, 9, 12?

- 10
- 2
- 3
- 7

Understanding the data

- Based on mean, median and mode, dataset may be classified as being negatively skewed, symmetrical, or positively skewed; and unimodal or bimodal.

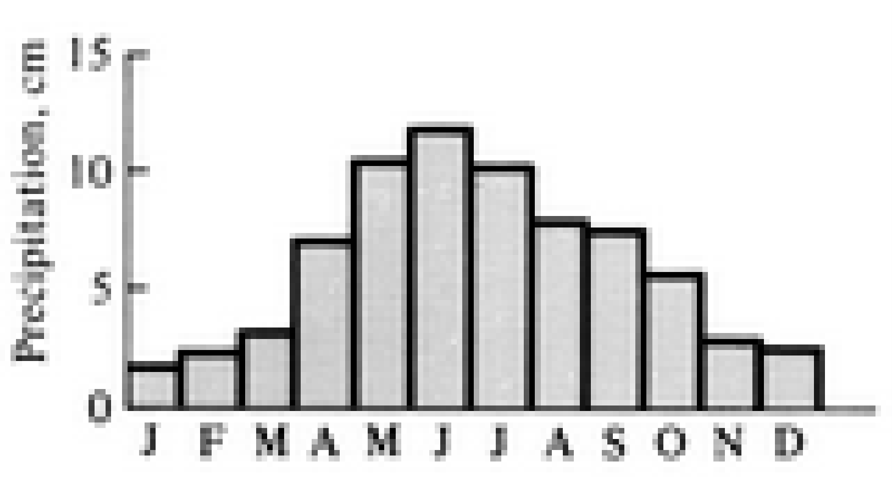


Measure of Location

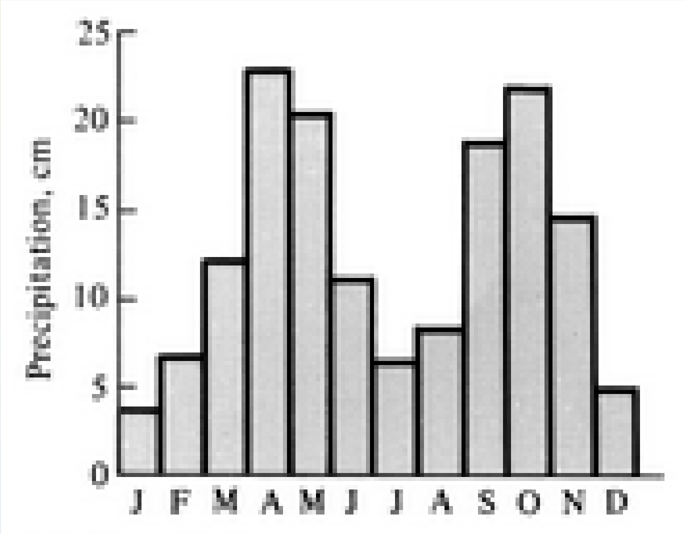
Understanding the data

- Based on mean, median and mode, dataset may be classified as being negatively skewed, symmetrical, or positively skewed; and unimodal or bimodal.

Unimodal distribution



Bimodal distribution



The terms specifically refer to uni/bi/trimodal as they relate to the **number of modes**.

However, a *bi* or *trimodal distribution* <u>could have 2 or 3 peaks with different heights</u>.
[You'd have to consider whether the peaks were prominent or not, and this is beyond the scope of this module.]

Measure of Location

Understanding the data: Discussion of the mean

- **Example:** Seven houses were sold last week in district X.
- Here are the selling prices : \$94,900, \$97,900, \$99,900, \$100,900, \$102,900, \$107,900, and \$1,250,000.
- The following is the calculated mean:

$$\begin{aligned}\mu &= \frac{\sum x}{N} \\ &= \frac{94{,}900 + 97{,}900 + 99{,}900 + 100{,}900 + 102{,}900 + 107{,}900 + 1{,}250{,}000}{7} \\ &= \frac{1{,}854{,}400}{7} \\ &\approx 264{,}914.29\end{aligned}$$

Understanding the data: Discussion of the mean (problems/solutions)

- **OUTLIERS**: The mean is highly influenced by extreme data values (outliers)
  - can be misleading for very skewed (positively or negatively) dataset.

- How do you propose we solve the MEAN issue?
  - Solution1: If you were to calculate the mean without the outlier, we would produce a value of approximately $100,733.33. This number (more) accurately describes the other six values.
  - Solution2: Another thing to try would be use the Median instead.

DESCRIPTIVE STATISTICS

MEASURE OF SPREAD

Measure of Spread

What is spread?

- Measure of location alone is insufficient to summarize a dataset
- Another numerical measure called measure of spread or variability is often also needed.
- Measures of spread is a measure of how far the data are spread out.

Example: Two different dataset having the same mean:

Given 1, 4 and 7	Given 3, 4 and 5
$\bar{x} = 4$	$\bar{x} = 4$

Measure of Spread

What is spread?

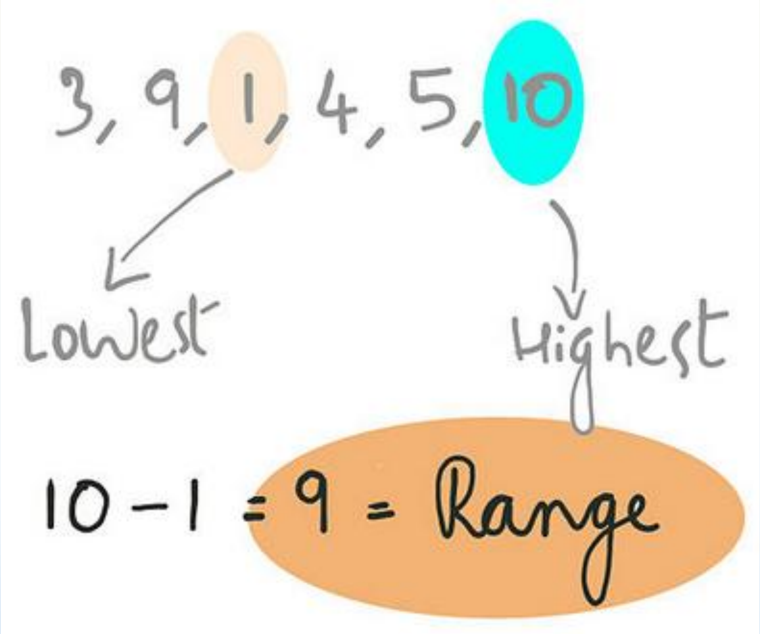
Quantitative Data

- Range
- Interquartile Range
- Variance
- Standard Deviation

Measure of Spread

Range

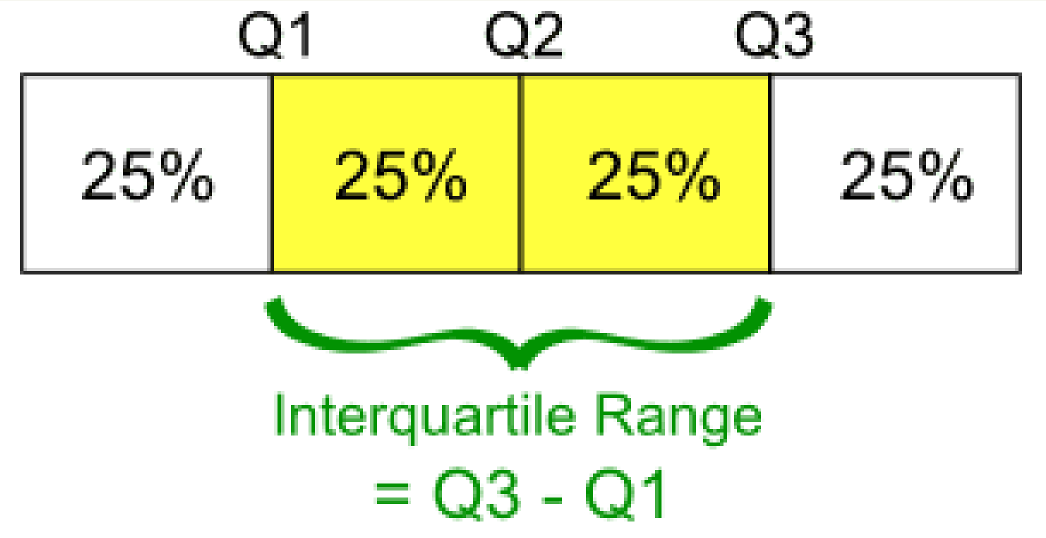
- Range is defined as the difference between the lowest and highest data values.
- Like mean, range can sometimes be misleading because it can be easily influenced by extreme data values (outliers).



Measure of Spread

Range: Interquartile range (IQR)

- Interquartile range (*IQR*) is defined as the difference between lower quartile *Q*1 (25%) and upper quartile *Q*3 (75%). Note that median is also known as middle quartile *Q*2 (50%).

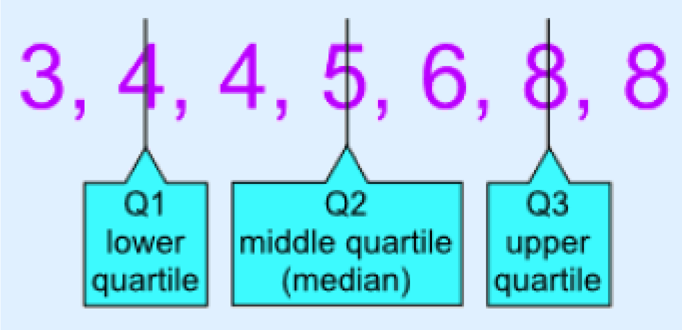


Measure of Spread

Range: Interquartile range (IQR): Example

- Calculate the interquartile range for the following
  - 3, 4, 4, 5, 6, 8,8

Example 1: odd-number of data



$$IQR = Q3 - Q1$$
$$= 8 - 4 = 4$$

Steps:

1. First sort the data (if not sorted)
2. Find the median (Q2), the value that splits the data into two groups.
3. Then find the median of the first group (Q1) and the median of the second group (Q3)

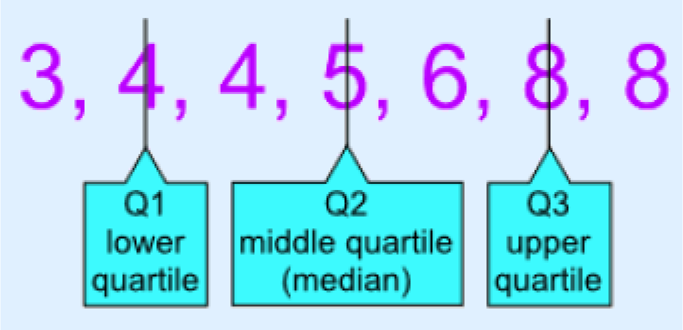
Measure of Spread

Range: Interquartile range (IQR): Example

- Calculate the interquartile range for the following

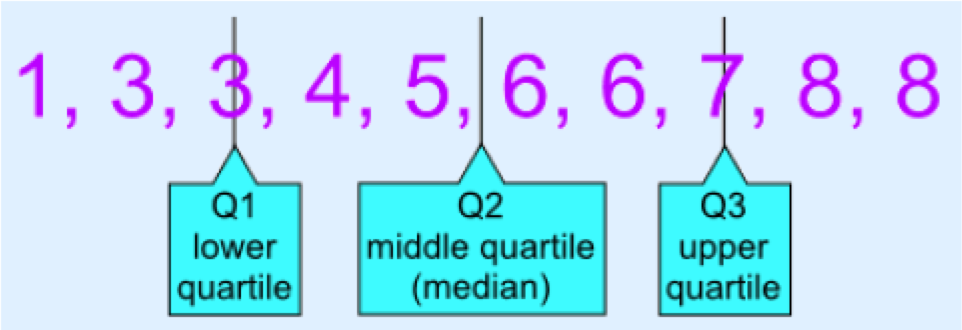
1, 3 , 3 , 4 , 5 , 6 , 6 , 7 , 8 , 8

Example 1: odd-number of data



$$IQR = Q3 - Q1$$
$$= 8 - 4 = 4$$

Example 2: even-number of data



$$IQR = Q3 - Q1$$
$$= 7 - 3 = 4$$

Measure of Spread

Range: Interquartile range (IQR): Example with Boxplot

- The lowest value, quartiles Q1, Q2, Q3 and highest value together form the so-called five-number summary and are commonly presented graphically in a Boxplot.

Example:

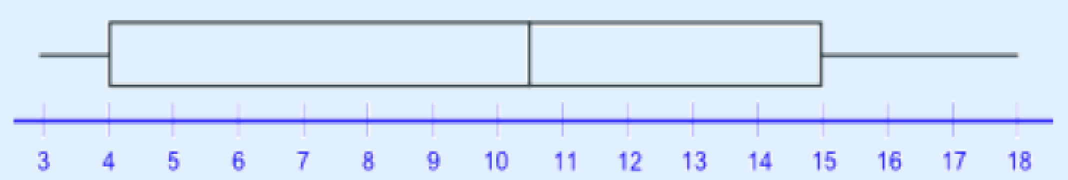
4, 17, 7, 14, 18, 12, 3, 16, 10, 4, 4, 11

Reorder and 'cut' into quartiles:

3, 4, 4 | 4, 7, 10 | 11, 12, 14 | 16, 17, 18

- The Lowest Value is **3**,
- The Highest Value is **18**

- Quartile 1 (Q1) = (4+4)/2 = **4**
- Quartile 2 (Q2) = (10+11)/2 = **10.5**
- Quartile 3 (Q3) = (14+16)/2 = **15**

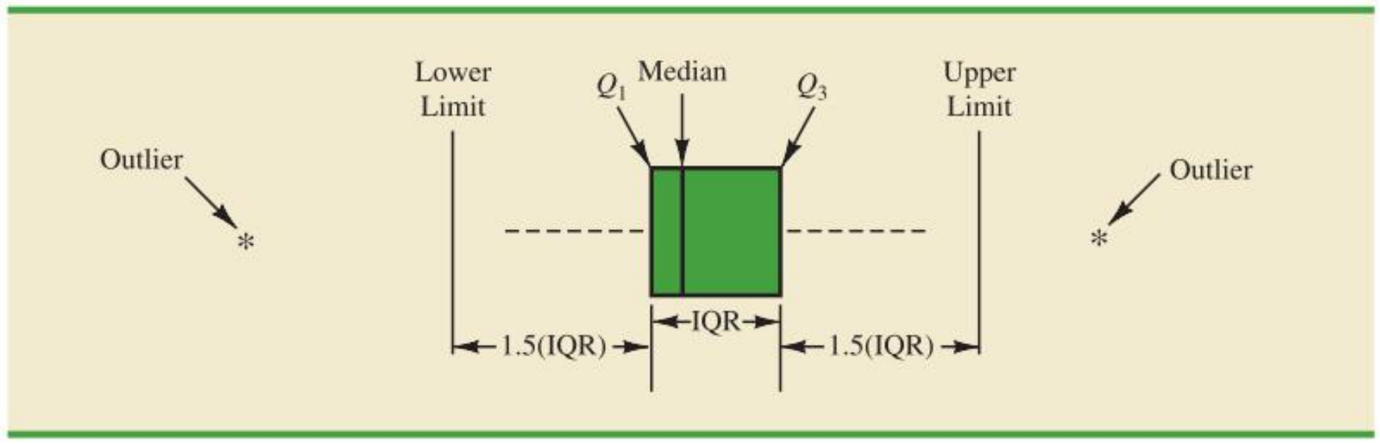


Measure of Spread

Range: Interquartile range (IQR): Example with Boxplot Outliers

- Boxplot is also commonly called the box and whisker plot and note that the whisker is not drawn beyond 1.5 x *IQR* on both sides.
- Any data value outside of lower or upper limit are defined as outliers and are represented by asterisk, circle or square.

Outlier = observation falling
below Q1 – 1.5(IQR)
or above Q3 + 1.5(IQR)



How to handle outliers?
- Legitimate data: keep it
- Error in measurement: discard or correct it

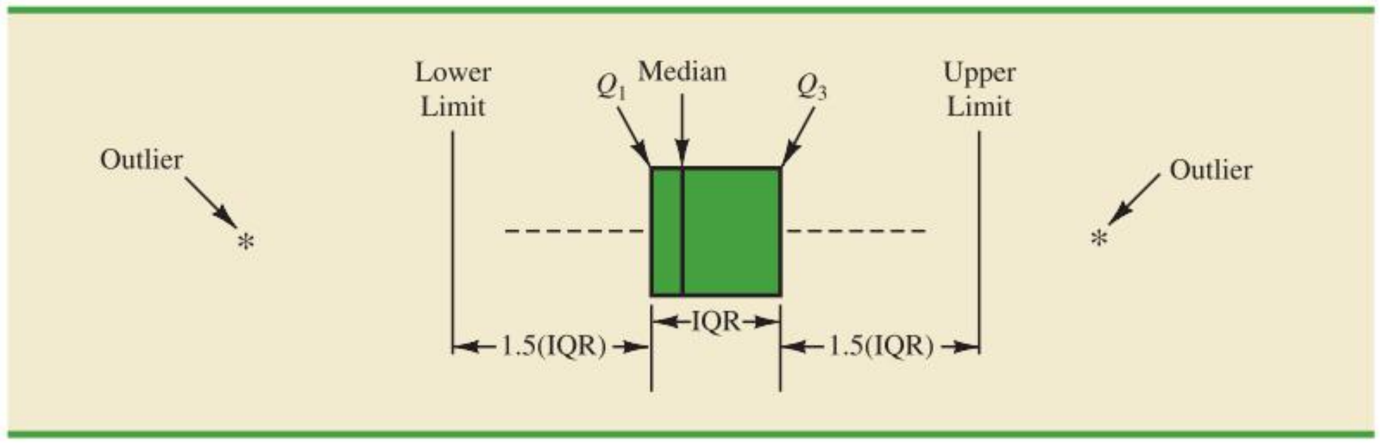
Measure of Spread

Range: Interquartile range (IQR): Example with Boxplot Outliers

- Boxplot is also commonly called the box and whisker plot and note that the whisker is not drawn beyond 1.5 x *IQR* on both sides.
- Any data value outside of lower or upper limit are defined as outliers and are represented by asterisk, circle or square.

Outlier = observation falling
below Q1 – 1.5(IQR)
or above Q3 + 1.5(IQR)

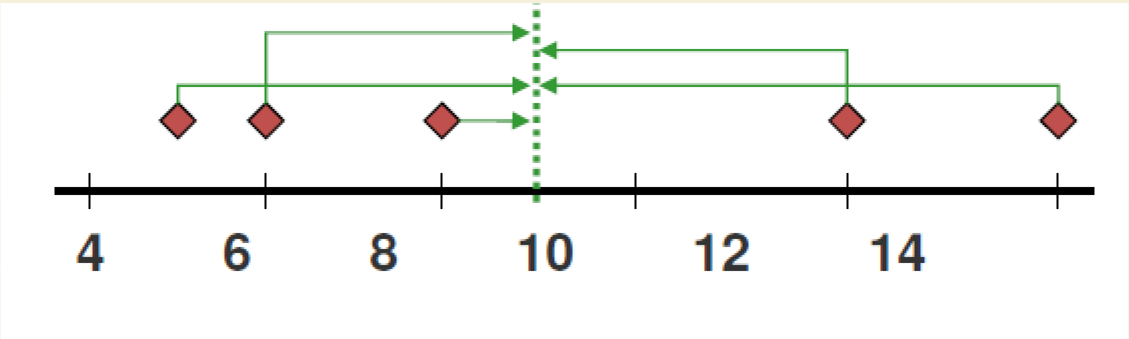
Ex. If Q1 = 2, Q3 = 10, IQR = 8
Outliers :
above 10 + 1.5(8) = 22
or **below** 2-1.5(8)=-10



Measure of Spread

Variance

Variance is defined as the average of the squared deviations ("distance") from the mean, where deviation is the difference between each data x_i and the mean.



Definition: Population variance, denoted by σ^2, is defined as (μ is the population mean):

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Definition: Sample, denoted by s^2, is defined as ($\bar{x}$ is the sample mean):

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

where $x_1, x_2, \ldots, x_n$ is a subset of $n < N$ data in the population.

Measure of Spread

Standard Deviation

- Standard deviation is related to variance and is defined as the square root of variance.

Definition: Population standard deviation, denoted as σ, is defined as:

$$\boxed{\sigma = \sqrt{\sigma^2}}$$

Sample standard deviation, denoted as s, is defined as:

$$\boxed{s = \sqrt{s^2}}$$

Measure of Spread

Variance & Stdev: Alternative formulae

To simplify calculations, variance is also commonly expressed in an alternative but equivalent formula as follows:

Alternative formula for population variance:

$$\boxed{\sigma^2 = \frac{1}{N}\left[\sum x_i^2 - \frac{\left(\sum x_i\right)^2}{N}\right]}$$

Alternative formula for sample variance:

$$\boxed{s^2 = \frac{1}{n-1}\left[\sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n}\right]}$$

Variance & Stdev: Example

- Example: Consider the sample data 5, 12, 6, 8 and 14.

- Calculate Variance and Standard Deviation

Variance & Stdev: Example (Sample Variance)

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Example: Consider the sample data 5, 12, 6, 8 and 14.

	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
	5	-4	16
	12	3	9
	6	-3	9
	8	-1	1
	14	5	25
Sum	45	0	60

$$\bar{x} = \frac{45}{5} = 9$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$s^2 = \frac{60}{5-1} = 15$$

$$s = \sqrt{15} = 3.87$$

Variance & Stdev: Example (Sample Variance) Alternative calculation

- Note the simplicity of using an alternative formula to compute variance and standard deviation:

Example: Consider the same sample data 5, 12, 6, 8 and 14.

	x_i	x_i^2
	5	25
	12	144
	6	36
	8	64
	14	196
Sum	45	465

$$s^2 = \frac{1}{n-1}\left[\sum x_i^2 - \frac{(\sum x_i)^2}{n}\right]$$

$$s^2 = \frac{1}{5-1}\left[465 - \frac{45^2}{5}\right] = 15$$

$$s = \sqrt{15} = 3.87$$

MEASURE OF SPREAD

VARIANCE & STANDARD DEVIATION FOR GROUPED DATA

Measure of Spread

Variance & standard deviation for grouped data

These formulae are fine when we have all our individual data points, but what if our data is grouped?

Standard deviation formulae

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Alternative formula for sample variance:

$$s^2 = \frac{1}{n-1}\left[\sum x_i^2 - \frac{(\sum x_i)^2}{n}\right]$$

Measure of Spread

Variance & standard deviation for grouped data

These formulae are fine when we have all our individual data points, but what if our data is grouped? We need to consider frequency (f_i)

$$s^2 = \frac{\sum f_i * (x_i - \bar{x})^2}{n-1}$$

$$s^2 = \frac{1}{n-1}\left(\sum f_i * x_i^2 - \frac{(\sum f_i * x_i)^2}{n}\right)$$

DESCRIPTIVE STATISTICS

COEFFICIENT OF VARIATION

Coefficient of Variation

Risk

- Lastly, a measure of relative variability called coefficient of variation is defined as the ratio of standard deviation over mean as follows:

Definition: The coefficient of variation is defined as:

$$CV = \frac{\sigma}{\mu}\ (\times 100\%)$$

or

$$CV = \frac{s}{\bar{x}}\ (\times 100\%)$$

- In practice, the coefficient of variation is often used as a measure of risk.
- The coefficient of variation is useful for comparing two or more variables having different means and different standard deviations.

Risk: Example

Example: Investing in which stock fund is riskier, Stock Fund A or B?

Stock Fund	Mean yearly return	Standard deviation
A	10.39%	16.18%
B	7.7%	13.82%

Definition: The coefficient of variation is defined as:

$$CV = \frac{\sigma}{\mu}\ (\times 100\%)$$ or $$CV = \frac{s}{\bar{x}}\ (\times 100\%)$$

Risk: Example

Example: Investing in which stock fund is riskier, Stock Fund A or B?

Stock Fund	Mean yearly return	Standard deviation
A	10.39%	16.18%
B	7.7%	13.82%

Solution:

Note that with different means, we cannot simply conclude that Stock Fund A is riskier because its standard deviation is larger than B.

Instead, we need to compute relative variability; i.e. coefficient of variation:

For Fund A: $$CV = \frac{16.18}{10.39} = 1.557$$

For Fund B: $$CV = \frac{13.82}{7.7} = 1.795$$

Definition: The coefficient of variation is defined as:

$$\boxed{CV = \frac{\sigma}{\mu}\ (\times 100\%)} \quad \text{or} \quad \boxed{CV = \frac{s}{\bar{x}}\ (\times 100\%)}$$

Since CV of B > CV of A, we can conclude that Fund B in fact has a higher investment risk than Fund A.

Coefficient of Variation

Risk

- Note: Summation of CV gererally occurs when assessing risk for multiple samples

Definition: The coefficient of variation is defined as:

$$\boxed{CV = \frac{\sigma}{\mu}\ (\times 100\%)} \quad \text{or} \quad \boxed{CV = \frac{s}{\bar{x}}\ (\times 100\%)}$$

- In which case, we summate the "*means*" (**μ**) and the "*standard deviations*" (**σ**)
- This indicates that the **more samples**, the less the risk

What is we’ve learnt?

Statistics: Descriptive and Inferential

Type of Data: Quantitative and Qualitative

Bivariate and contingency tables

Frequency Tables, relative frequency table

Graphical representation: Bar charts and histograms and boxplots

Measure of location: Mean, Median, Mode

Measure of spread: Range, IQR, variance, standard deviation and coefficient of variation